

Siddiqui et al. 2022 — TMCMC Re-calibration Report

10,000-particle GPU-accelerated Bayesian Inference
on Independent External Dataset (Siddiqui et al. 2022, PMC8828709)

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1 Summary

TMCMC re-calibration of the 5-species Hamilton ODE model on Siddiqui et al. (2022, *Dental Materials*, DOI: 10.1016/j.dental.2021.12.021, PMC8828709) in vitro biofilm data. Six species (*S. oralis*, *A. naeslundii*, *A. actinomycetemcomitans*, *V. parvula*, *F. nucleatum*, *P. gingivalis*) were co-cultured on polished cpTi disks and monitored over 21 days (6h, 1, 3, 7, 14, 21d). *A. actinomycetemcomitans* was excluded and data renormalized to 5 species. Two fractions reported: planktonic (bulk medium) and adherent (biofilm on cpTi surface).

Key differences from Heine 2025 (our calibration data):

- No antibiotic or antimicrobial treatment—pure succession
- Single culture condition (no commensal/dysbiotic manipulation)
- Substrate: polished cpTi (dental implant) vs hydroxyapatite (Heine)
- Species: *V. parvula* (SM) vs *V. dispar* (Heine)—same genus, interchangeable in oral models
- Inoculum: So monoculture (Day 0.25 $\approx$ 99% So) vs mixed inoculum (Heine)

Table 1: Fit quality (10,000 particles, RW $\times$ 10 mutation steps, RTX 4090 GPU)

Dataset	RMSE	Cosine sim	Spearman ρ	max log L
Planktonic	0.3260	0.5484	0.5831	-20.6
Adherent	0.1387	0.9243	0.7948	-15.3

Key findings:

- Adherent dataset shows excellent fit (Cos. sim 0.924, RMSE 0.139)
- Planktonic has rapid Fn surge (Day 3: 74%) that is harder to capture
- Day 21 planktonic: $E = 469$ Pa; adherent: $E = 462$ Pa
- Both within literature range [20, 14000] Pa

2 Posterior Predictive: Species Succession (5-panel)

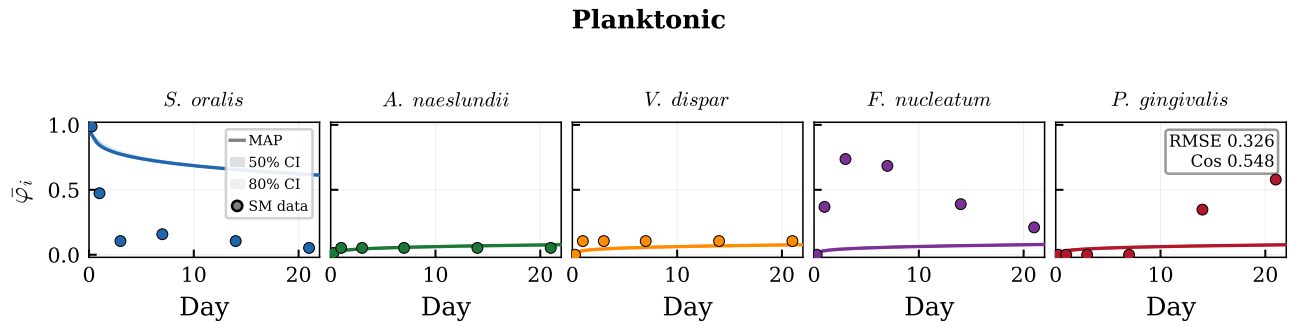


Figure 1: Planktonic: MAP trajectory (line), 50%/80% CI (shaded), SM data (dots).

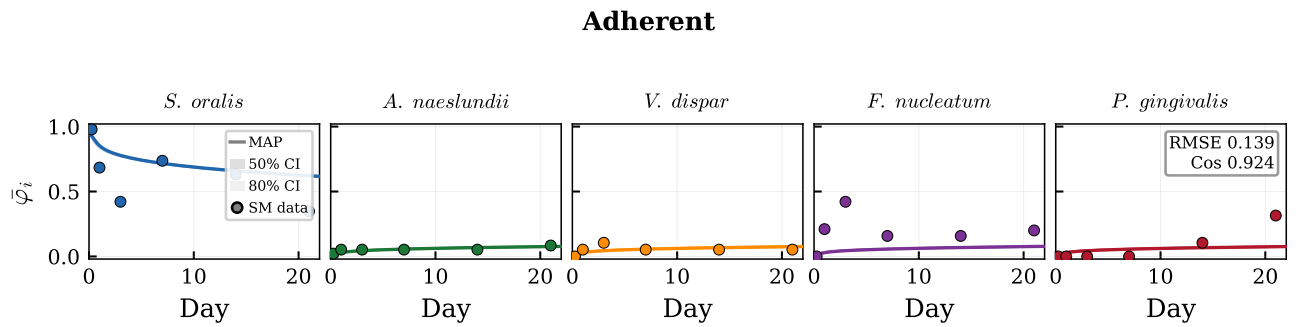


Figure 2: Adherent on polished cpTi.

3 Interaction Matrix A

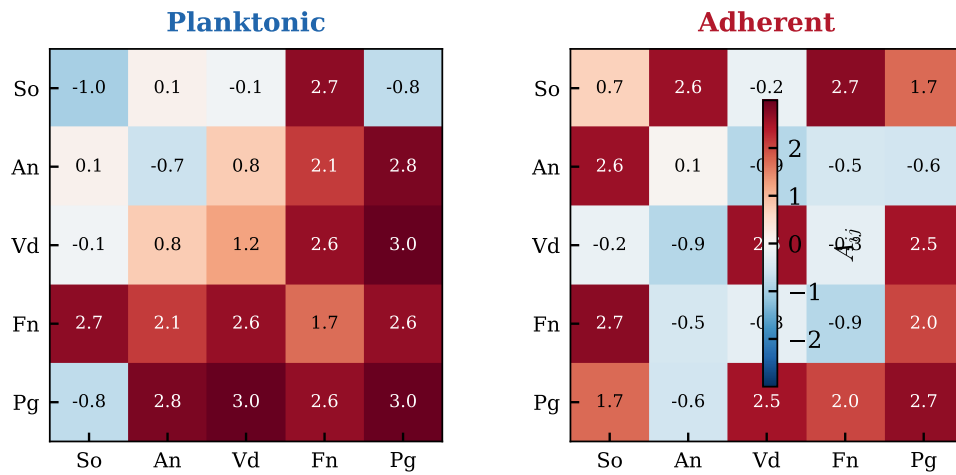


Figure 3: MAP interaction matrices. Red = mutualistic, blue = competitive. Note planktonic has strong Fn–Pg coupling ($a_{45} = 2.6$) vs adherent ($a_{45} = 2.0$).

4 Growth Rates μ_i

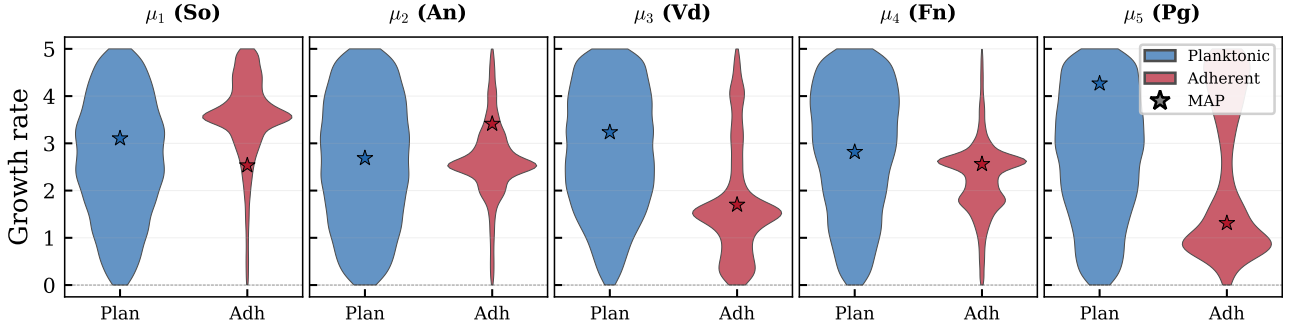


Figure 4: Posterior distribution of growth rates. Star = MAP. Pg growth highest in planktonic ($\mu_5 = 4.27$) driving Day 21 dominance.

5 Dysbiosis Index and Young's Modulus

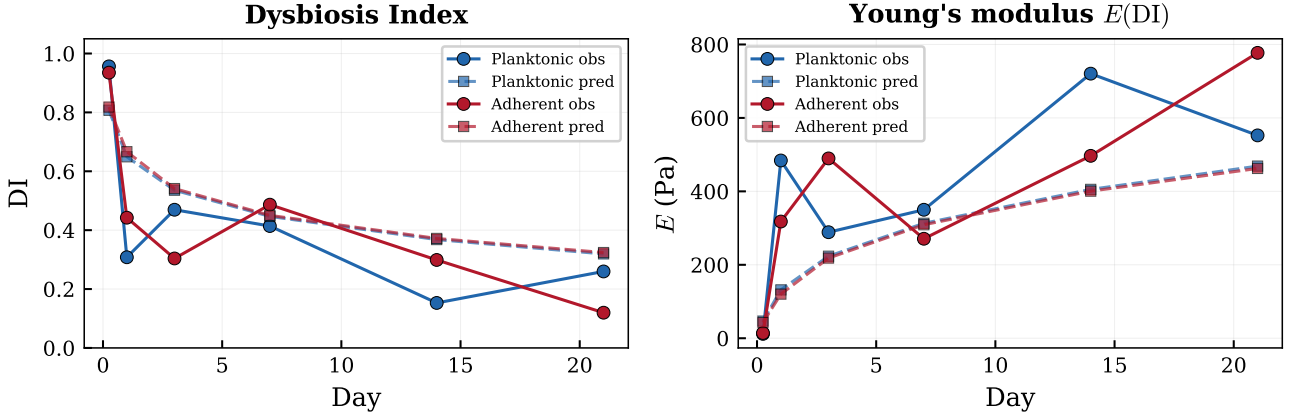


Figure 5: DI and $E(DI)$ trajectories.

6 Observed vs Predicted

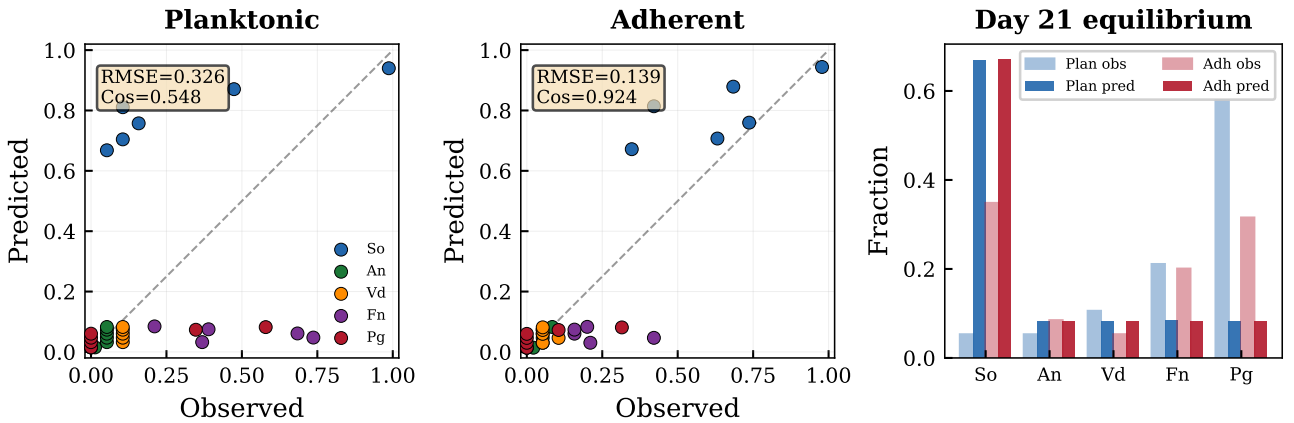


Figure 6: Left/center: scatter. Right: Day 21 species composition.

7 Full Posterior (20 parameters)

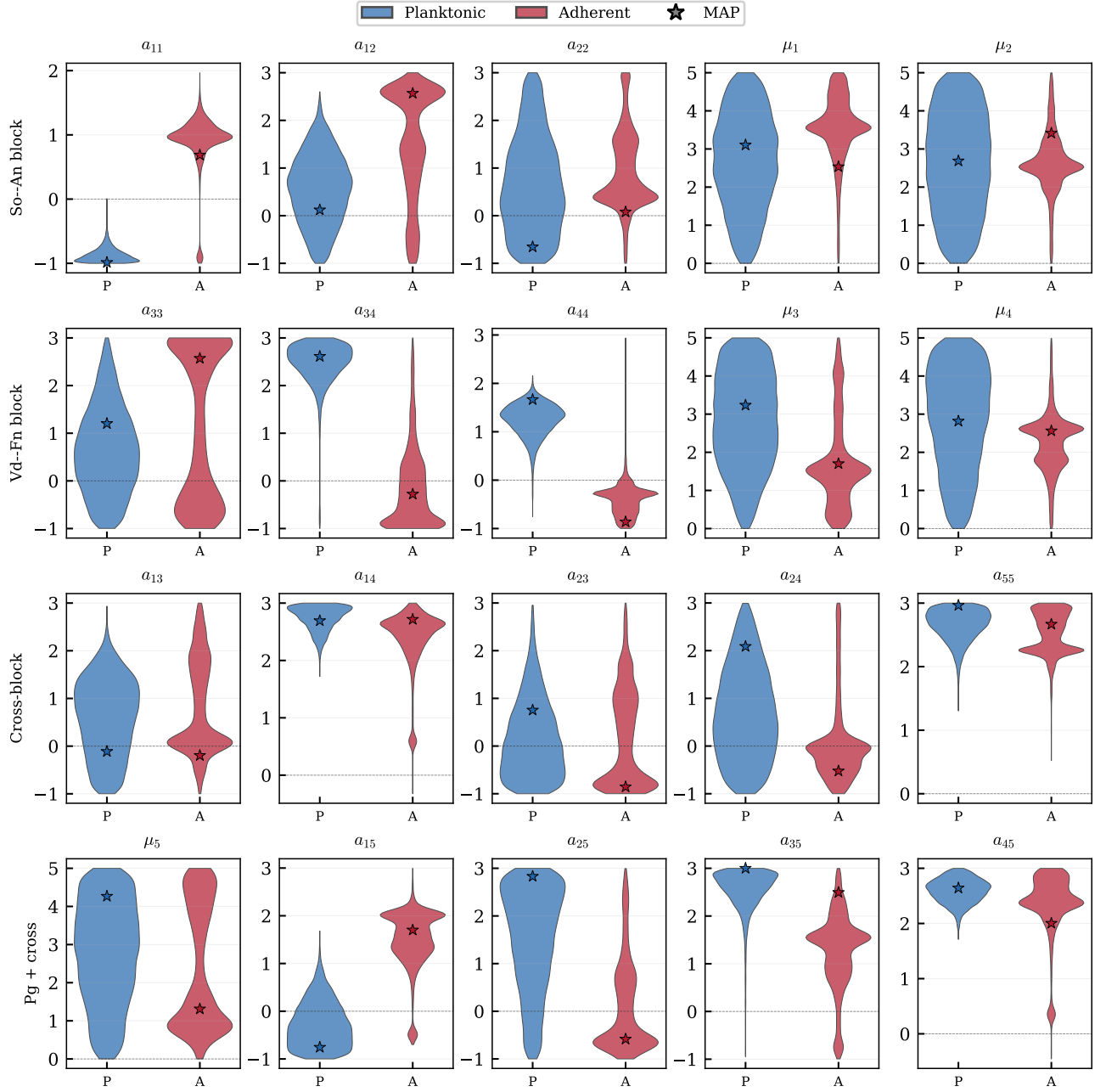


Figure 7: All 20 parameters: interaction (a_{ij}) and growth (μ_i). P = Planktonic, A = Adherent.

8 MAP Parameters

Table 2: MAP θ (10,000 particles)

Idx	Param	Planktonic	Adherent
0	a_{11}	-0.987	+0.686
1	a_{12}	+0.121	+2.568
2	a_{22}	-0.655	+0.078
3	μ_1	+3.105	+2.533
4	μ_2	+2.686	+3.413
5	a_{33}	+1.203	+2.570
6	a_{34}	+2.612	-0.278
7	a_{44}	+1.664	-0.863
8	μ_3	+3.234	+1.700
9	μ_4	+2.817	+2.561
10	a_{13}	-0.112	-0.197
11	a_{14}	+2.693	+2.716
12	a_{23}	+0.753	-0.856
13	a_{24}	+2.088	-0.524
14	a_{55}	+2.964	+2.668
15	μ_5	+4.266	+1.311
16	a_{15}	-0.756	+1.701
17	a_{25}	+2.831	-0.585
18	a_{35}	+3.000	+2.493
19	a_{45}	+2.643	+2.003

9 Planktonic Accuracy Improvement Ideas

Current planktonic RMSE (0.326) is higher than adherent (0.139). Root causes and potential fixes:

1. **Rapid Fn surge (Day 1–3):** Fn jumps 0→74% in 3 days. The Hamilton ODE with uniform Δt struggles to resolve this. *Fix:* Adaptive time-stepping or finer Δt near early phase.
2. **Hill gate delay:** Current $K = 0.05$, $n = 2$ may be too restrictive for SM where Pg appears by Day 14. *Fix:* Include K_{hill} and n_{hill} as free parameters in TMCMC (2 extra dims, 22 total).
3. **So monoculture IC:** Day 0.25 is 99% So. This sharp IC amplifies early-phase sensitivity. *Fix:* Use Day 1 as IC (skip Day 0.25) to avoid the transient.
4. **Nutrient coupling:** Planktonic cells compete for nutrients in bulk media (well-mixed). Adding Monod nutrient ODE (`simulate_0d_nutrient`) could explain the Fn–So switch.
5. **Prior tightening:** Wide prior $[-1, 3]$ may allow multimodal posterior. *Fix:* Use Heine 2025 posterior as informative prior (hierarchical Bayes).
6. **σ_{obs} schedule:** Currently fixed at 0.10. Early timepoints (digitized from bar charts) have higher uncertainty. *Fix:* Per-timepoint $\sigma_i = [0.15, 0.12, 0.10, 0.08, 0.06, 0.05]$.

10 Configuration

- **Particles:** 10,000 (vmap GPU parallel)
- **Mutation:** RW $\times$ 10 steps/particle/stage
- **σ_{obs} :** 0.10

- **ODE:** 2,500 steps ($\Delta t = 10^{-4}$), $K = 0.05$, $n = 2$
- **Prior:** $a_{ij} \in [-1, 3]$, $\mu_i \in [0, 5]$
- **GPU:** NVIDIA RTX 4090 (vancouver01)